In this type of iron box, according to the type of cloth, the heat can be controlled manually, by connecting or disconnecting the supply through switch.

3.2.5 Automatic iron box

In an automatic iron box, the temperature of the cloths can be selected to the required quantity of heat. The temperature is controlled by the thermostat in the appliance automatically. Hence this appliance is called as Automatic Iron Box.



Fig. 3.2 Electric automatic iron box

a) Construction

As shown in figure 3.2, the construction and operation of automatic iron box is similar to that of an Ordinary iron box. In an automatic iron box, the temperature is controlled with the help of thermostat. The thermostat is a bi-metallic strip. In addition to this, an indicating lamp is connected in series with the heating element.

b) Indicating lamp

In the iron box LED lamp is used as an indicating lamp. Indicating lamp is provided to know whether the supply is going into the appliance or not. During the time of supply, the indicating lamp glows. After attaining the fixed level in the regulating knob (or) Selector knob the indicating lamp turns off automatically. This shows the availability of supply in the appliance.

c) Thermostat

Thermostat is a bi-metallic strip made up of two different metals. Generally, due to heat, the metals get expanded. After setting the required value in the regulating knob, the heat increases gradually and gets saturated. Then the bi-metallic strip will bend automatically and to stop the supply to the heating element. This can be viewed in the figure 3.3.

After the heat gets reduced in the sole plate, the bi-metallic strip, again contacts with electric supply and makes the appliance to get heated as shown in figure 3.4. The expansion of the strip will vary, according to the strip which the bi-metal was made.

d) Working principle of automatic iron box

The electric input is given to the iron box by putting the switch in ON position. The current goes to the heating element of the iron box through power chord. The heating element gets heated up due to its heat production property. The heat was absorbed by the sole plate and gets heated gradually. This heat removes the wrinkles in the clothes.

3.2.6 Steam iron box

The construction and working principle of a steam iron box is similar to

BEE_ENG_Chapter_03.indd 40 12/27/2019 10:24:51 AM